Chapter 02 : Literature Survey

2.1 Overview of Relevant Literature

The aims of our literature review were to review the most recent advances made from 2022 to 2025 with AI-enabled imaging within the field of healthcare, specifically focusing on the development of image quality improvements, anomaly detection and explainable artificial intelligence (AI). The most prominent trend during this period has been the notable transition from traditional CNN (convolutional neural network) based architectures to more advanced transformer architectures that utilize an attention mechanism, enabling models to gain a more nuanced understanding of images. Major advances include improved deep learning methods to denoise images, improved U-net architectures with better abilities for the accurate segmentation of images, and newly developed methods, such as Grad-CAM, that can assist medical professionals in understanding the output of the AI model so that they can be more confident when using these technologies in clinical practice. Although the models described above generally achieve very high levels of accuracy, typically over 95%, there are also several challenges that need to be addressed including increasing the computational power required to run these models, developing new broad-based datasets for training models and developing generalizable models so that the models can be used across all image types seamlessly.

Table 2.1 Literature Review

S. No	Author & Paper Title	Year	Tools / Techniques / Dataset	Key Findings / Results	Limitations / Gaps Identified
1	Anuj Kumar, Satya Prakash Yadav, Awadhesh Kumar – *An improved feature extraction algorithm for robust Swin Transformer*	2025	CNN + Swin Transformer; CT & MRI datasets	Higher accuracy, robust to noise, less computation while maintaining performance	Needs strong hardware, mainly tested on medical images, untested on adversarial attacks or poor-quality data
2	Shoffan Saifullah, Rafal Drezewski, Anton	2025	CNNs + Transformers;	High accuracy, detects diverse	High computation,

	Yudhana, Andiko Putro Suryotomo – *Automatic Brain Tumor Segmentation: Advancing U-Net With ResNet50 Encoder*		BraTS dataset; Dice, sensitivity, accuracy	tumors, robust across patients, speeds up diagnosis	large datasets needed, struggles with small tumors or artifacts, limited generalization beyond BraTS
3	Zhe Guo, Xiang Li, Heng Huang, Ning Guo, Quanzheng Li – *Deep Learning-Based Image Segmentation on Multimodal Medical Imaging*	2025	CNNs + Random Forest; STS dataset (50 patients, PET/CT/T1/T2 MRI); Dice	Multimodal fusion improved segmentation over single-modality networks	Single dataset, needs accurate registration, drops under extreme noise, only supervised CNN/U-Net used
4	Arghya Pal, Sailaja Rajanala, CheeMing Ting, Raphael Phan – *Denoising via Repainting: An image denoising method using layer-wise medical image repainting*	2025	Multi-scale anisotropic Gaussian blur + Bezier-path; MSE & Xing-loss; MRI datasets	Outperforms baseline denoisers in PSNR/SSIM, preserves edges and textures	Computationally heavy, parameter tuning required, limited to 2D images
5	Hongxu Jiang, Muhammad Imran, Teng Zhang, Yuyin Zhou, Muxuan Liang, Kuang Gong, Wei Shao – *Fast-DDPM: Fast Denoising Diffusion Probabilistic Models for Medical Image-to-Image Generation*	2025	Fast-DDPM, U-Net, conditional diffusion, task-specific noise schedulers	Outperforms GANs/CNNs/standard diffusion in PSNR/SSIM and visual quality; reduces training/inference time	2D models only, manual noise schedulers; 3D modeling and adaptive scheduling future work
6	Leila Khaertdinova, Tatyana Shmykova, Ilya Pershin, et al. –	2025	WORD CT dataset, Tobii Eye Tracker,	Eye-gaze corrections improved	Single dataset, 2D slices, specialized

	Gaze Assistance for Efficient Segmentation Correction of Medical Images		MedSAM, AdamW; Dice + cross-entropy	segmentation for small/complex organs; outperformed bounding boxes	hardware, small organs still challenging
7	Narayan Hampiholi – *Medical Imaging Enhancement with AI Models for Automatic Disease Detection & Classification*	2025	RNNs (sequential), 3D CNNs, multimodal AI	Improved disease detection, early diagnosis, precise tumor localization, reduced false positives	Data bias, poor generalization, ethical/privacy concerns, reliance on human oversight
8	Saif Ur Rehman Khan, Sohaib Asif, Ming Zhao, et al. – *Optimized deep learning model for comprehensive medical image analysis across multiple modalities*	2025	MobileNet (truncated), ConvLSTM, modified Inception, metaheuristic optimization, Grad-CAM	Accuracy 97.37%, lightweight (2.16M parameters), robust multi-disease detection, interpretable	Large labeled data needed, untested on new modalities, lower accuracy for complex diseases
9	Ravi Ranjan Kumar, Rahul Priyadarshi – *Denoising and segmentation in medical image analysis: A comprehensive review on ML and DL approaches*	2025	CNNs, ConvLSTM, MobileNet, inception blocks, metaheuristic optimization, Grad-CAM	97.37% accuracy, high precision/recall, 2.16M parameters, 6s test time	Needs large datasets, lower accuracy on complex cases, untested on new modalities
10	Xiang Li, Like Li, Yuchen Jiang, et al. – *Vision-Language Models in medical image analysis: From*	2025	VLMs, self-supervised learning, multi-modality fusion, knowledge graphs,	Better diagnosis, interpretable outputs, privacy protection, faster processing	Data scarcity, alignment issues, low interpretability, bias, high computation

	simple fusion to general large models		federated/cloud computing		
11	Bin Yu, Quan Zhou, Li Yuan, et al. – *3D medical image segmentation using serial-parallel CNN and transformer with cross-window self-attention*	2024	MRI/CT datasets, AdamW, MindSpore Lite; DSC/IoU/HD95/ASD	SPCT outperformed U-Net & Swin-UNETR; DSC 92%+ on ProstateX & Atrium, 81% on Pancreas; efficient computation	High GPU memory needed, tested on single-organ tasks, detail loss with 4× upsampling
12	Johnson Oyeniyi, Paul Oluwaseyi – *Emerging Trends in AI-Powered Medical Imaging: Enhancing Diagnostic Accuracy and Treatment Decisions*	2024	CNNs, RNNs, 3D CNNs, transformers, multimodal fusion, federated learning	Improved accuracy, faster analysis, personalized care	Needs large labeled datasets, high computation, bias & privacy/security risks
13	Mohamed Musthafa M, Mahesh T. R, Vinoth Kumar V, Suresh Guluwadit – *Enhancing brain tumor detection in MRI using explainable AI with Grad-CAM & ResNet50*	2024	ResNet50, transfer learning, PyTorch, data augmentation, Grad-CAM	Test accuracy 98.52%, precision & recall 97–99%	Small, demographically limited dataset; may reduce generalizability
14	Muhammad Aasem, Muhammad Javed Iqbal – *Toward explainable AI in radiology: Ensemble-CAM for thoracic disease localization in chest X-ray*	2024	Deep CNNs, Ensemble-CAM; chest X-ray datasets	High accuracy (up to 98%), better localization than single models	High computation, weaker than fully supervised methods, dataset quality dependent

15	Tahir Hussain, Hayaru Shouno – *MAGRes-UNet: Improved Medical Image Segmentation Through Multi-Attention Gated Residual U-Net*	2024	MAGRes-UNet with attention gates, AdamW; BT MRI & HAM10000 datasets	Accuracy 99.8–99.9%, Dice/IoU ~97%, high sensitivity & specificity	2D model lacks 3D context; brain tumor segmentation challenging due to MRI complexity & limited labels
16	Yun Zi, Qi Wang, Zijun Gao, et al. – *Application of Deep Learning in Medical Image Segmentation & 3D Reconstruction*	2024	Self-attention U-Net, cross-entropy loss, PyTorch	MIS Dice 85.3%, IoU 78.9%; 3D reconstruction MSE 0.015, SSIM 0.94	2D model misses full 3D context; sensitive to hyperparameters ; needs better generalization
17	Fabian Isensee, Tassilo Wald, Constantin Ulrich, et al. – *nnU-Net Revisited: A Call for Rigorous Validation in 3D Medical Image Segmentation*	2024	CNN/Transformer models (nnU-Net, Auto3DSeg); 6 3D medical datasets; DSC/NSD	CNN U-Nets outperform Transformers; scaling boosts performance on harder datasets	Most new methods don't surpass nnU-Net; Transformers need more data; some datasets saturate
18	Subhashis Suara, Aayush Jhat, Pratik Sinha, Arif Ahmed Sekh – *Is Grad-CAM Explainable in Medical Images?*	2023	Densenet169, Grad-CAM, data augmentation	96.7% accuracy; Grad-CAM highlights tumor regions, aiding interpretation	Needs better visualization; expand to other modalities
19	João Guerrero, Pedro Tomás, Nuno Garcia, Helena Aidos – *Super-resolution of MRI using GANs*	2023	FastMRI dataset, GANs, denoising (NLM, BM3D); MSE, PSNR, SSIM	SRResCycGAN best quantitatively; Beby-GAN & RankSRGAN visually accurate	GANs may produce artifacts; metrics may mislead; bicubic downsampling differs from real

					MRI; generalization limited
20	Waqar Ahmad, Hazrat Alias, Zubair Shah, Shoaib Azmat – *A new GAN for medical image super resolution*	2022	Medical imaging datasets (DRIVE, STARE, ISIC, BraTS); LR images from downsampled HR	Outperformed SRGAN & bicubic; PSNR +3–11 dB, SSIM +2–7%, robust to noisy images	High computation; generalization issues on unseen data

2.2 Key Gaps in the Literature

Several deficiencies are present in the literature regarding the application of AI and advanced deep-learning developments in patient imaging. One major obstacle to the successful deployment of these technologies is their reliance upon few, restricted datasets that represent only specific imaging modalities and/or types, which prevents them from properly adapting to clinical situations other than the one in which they were developed. Consequently, many of the currently available AI systems will likely struggle to perform correctly when presented with real-world scenarios such as blurred images, low resolution scans, or unexpected variations in an image from what is expected. In addition to this issue of data quality, the cutting-edge nature of currently available methodologies like transformers or GANs, and even diffusion models, requires significant amounts of computational power and extended amounts of time for training and tuning, which can present logistical complexities when trying to implement these technologies in an actual clinic.

Another issue is that the majority of the available studies are based predominantly on 2D images or slices; therefore, the critical aspect of spatial context (which exists only when viewing a structure in 3D) is being ignored. In addition, most models tend to focus only on isolated organs or tasks, making them significantly less useful for performing comprehensive studies of patients or for multi-organ evaluations. Generalization is another challenge for the majority of AI models; many will show significant reductions in performance when tested against new datasets and across different types of imaging, and the majority will be particularly sensitive to artifacts, or small lesions that go undetected. One of the remaining challenges is Interpretability. A variety of tools have been proposed by the research community to provide some insight into the inner workings of these models (e.g. Grad-CAM, vision-language setup), but many of these tools do not go as deep into interpretability as clinicians would need to feel completely assured. There are also other ethical considerations inherent to the use of these models, including the potential for demographic bias, and also the potential for under-utilization of techniques that improve upon patient privacy.

When examining the specific classes of clinical use cases that these models have been applied to, segmentation still encounters difficulties due to the complexity of organs, as well as intricate anatomical structures. Similarly, enhancing various areas of imaging with advances in super-resolution or noise reduction, using trending techniques such as GANs or Diffusion Models, may inadvertently lead to new forms of errors. Although the lab-bench performance of many of these models has been verified to be highly accurate, translating that accuracy into routine use within the clinical setting will require significant advances beyond what currently exist—namely, models that scale easily, generalize across multiple modalities, and provide information that clinicians can rely upon to inform their decisions.